Introduction to Programming

Quiz 2

Date: 26 October 2026 Time: 11:45 to 12:15 hrs Marks: 25

Name: _____ Roll no.: _____

- Part A has eight multiple-choice questions of 2 marks each. Circle (or tick) exactly one option.
- Part B has one program-tracing questions worth 9. Show your working.
- Use the lecture definitions unless stated otherwise.
- Use your own paper (or page 4) for rough work. Clearly mark the rough work portion.
- Write your name and roll number on top.

Part A: Multiple choice

$8 \times 2 = 16$ *marks*

Questions 1–3 use these declarations:

```
data Day = Sun | Mon | Tue | Wed | Thu | Fri | Sat
  deriving (Eq, Show, Ord, Enum)

data Shape = Circle Double | Square Double | Rectangle Double Double
  deriving (Eq, Ord, Show)
```

1. What does `[d | d <- [Sun ..], d > Thu]` evaluate to? [2]

(A) `[Fri,Sat]`
(B) `[Thu,Fri,Sat]`
(C) `[Sun,Mon,Tue,Wed]`
(D) Compile-time error: `Day` is not a numeric type, so `[Sun ..]` is not allowed

2. Which of the following evaluates to `True`? [2]

(A) `Square 1.0 < Circle 10.0`
(B) `Rectangle 1.0 5.0 < Rectangle 1.0 2.0`
(C) `Circle 9.0 < Square 1.0`
(D) `Square 2.0 > Rectangle 1.0 1.0`

3. What is the type of `map Rectangle [1.0, 2.0]`? [2]

(A) `[Shape]`
(B) `[Double -> Shape]`
(C) `Double -> [Shape]`
(D) Compile-time error: `Rectangle` needs two arguments

4. What is the most general type of `f`? [2]

```
f x ys = length (filter (== x) ys) > 0
```

(A) `a -> [a] -> Bool`
(B) `Ord a => a -> [a] -> Bool`
(C) `(Eq a, Num a) => a -> [a] -> Bool`
(D) `Eq a => a -> [a] -> Bool`

5. Consider the following code. [2]

```
data Coin = H | T deriving (Eq, Show)

instance Show Coin where
  show H = "Heads"
  show T = "Tails"
```

What happens when we load this file and evaluate `show [H, T]`?

(A) `"[Heads,Tails]"`
(B) `"[H,T]"`
(C) `"HeadsTails"`
(D) Compile-time error: duplicate `Show Coin` instance

6. Which expression is rejected by the type checker? [2]

(A) `elem (1, 'a') [(1, 'a'), (2, 'b')]`
(B) `max "abc" "abd"`
(C) `[(+3)] == [(*5)]`
(D) `read "42" + (1 :: Int)`

7. What happens when this definition is compiled? [2]

```
average :: [Double] -> Double
average xs = sum xs / length xs
```

(A) Compile-time error: `length xs` has type `Int`, but `/` needs both operands to have type `Double`
(B) It compiles, and `average [1, 2, 3]` returns `2.0`
(C) Compile-time error: `Double` is not an instance of `Num`
(D) It compiles, but `average []` crashes with a divide-by-zero exception

8. Each expression is typed at the `ghci` prompt. Which one *never finishes* printing its result? [2]

(A) `length (map (`div` 0) [0..9])`
(B) `take 3 (filter even [1..])`
(C) `head (2 : reverse [1..])`
(D) `filter (< 5) [1..]`

Part B: Program tracing *9 marks*

9. Consider the following code.

```
pick :: Int -> [Int] -> Int
pick _ []     = 0
pick 0 (x:_)  = x
pick n (_:xs) = pick (n-1) xs

nums :: [Int]
nums = map (*3) [1 ..]
```

(a) Show the leftmost-outermost reduction of `pick (4-2) nums` until no further simplification is possible. You may combine steps that unfold `map` and `[1 ..]` into a single step.

[5]

(b) What do the following return? Justify each answer in one line.

(i) `pick 1 [5 `div` 0, 7, 8]` [4]

(ii) `pick 1 [5, 7 `div` 0, 8]`